Practical 12: Event Handling with Listeners

Source: MODULE 2.pdf | Source pages: 26–29

Aim

To develop and test an Android application demonstrating event handling with listeners.

Steps to Perform the Practical

1. Create a project named EventHandlingDemo.

2. Add Click Me and Long Press Me buttons in activity_main.xml.

3. In MainActivity.java, attach an OnClickListener to the first button and an OnLongClickListener to the second.

4. Show a Toast when each listener is triggered. Return true from the long-click listener.

5. Run the app and tap the first button.

6. Long-press the second button and verify its Toast message.

Result

The application is run and the expected behavior is verified according to the practical instructions and code.

Note: Use the package name and resource names from the source code consistently. If Android Studio reports a missing resource or class, check spelling and ensure all files are in the correct folders.

5. Develop an Android application demonstrating Event Handling using Listeners (OnClick, OnLongClick, etc.).

Project Name: EventHandlingDemo

File Name: activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>

<LinearLayout

   xmlns:android="http://schemas.android.com/apk/res/android"

   android:layout_width="match_parent"

   android:layout_height="match_parent"

   android:orientation="vertical"

   android:gravity="center"

   android:padding="20dp">

   <Button

       android:id="@+id/btnClick"

       android:layout_width="wrap_content"

       android:layout_height="wrap_content"

       android:text="Click Me"/>

   <Button

       android:id="@+id/btnLongClick"

       android:layout_width="wrap_content"

       android:layout_height="wrap_content"

       android:text="Long Press Me"

       android:layout_marginTop="20dp"/>

</LinearLayout>
```

File Name: MainActivity.java

```
package com.example.eventhandlingdemo;


import android.os.Bundle;

import android.widget.Button;

import android.widget.Toast;

import androidx.appcompat.app.AppCompatActivity;


public class MainActivity extends AppCompatActivity {


   Button btnClick, btnLongClick;


   @Override

   protected void onCreate(Bundle savedInstanceState) {

       super.onCreate(savedInstanceState);

       setContentView(R.layout.activity_main);


       btnClick = findViewById(R.id.btnClick);

       btnLongClick = findViewById(R.id.btnLongClick);


       // OnClickListener

       btnClick.setOnClickListener(v -> {

           Toast.makeText(

                   MainActivity.this,

                   "Button Clicked",

                   Toast.LENGTH_SHORT
```

```
            ).show();

        });

        // OnLongClickListener

        btnLongClick.setOnLongClickListener(v -> {

            Toast.makeText(

                    MainActivity.this,

                    "Button Long Pressed",

                    Toast.LENGTH_SHORT

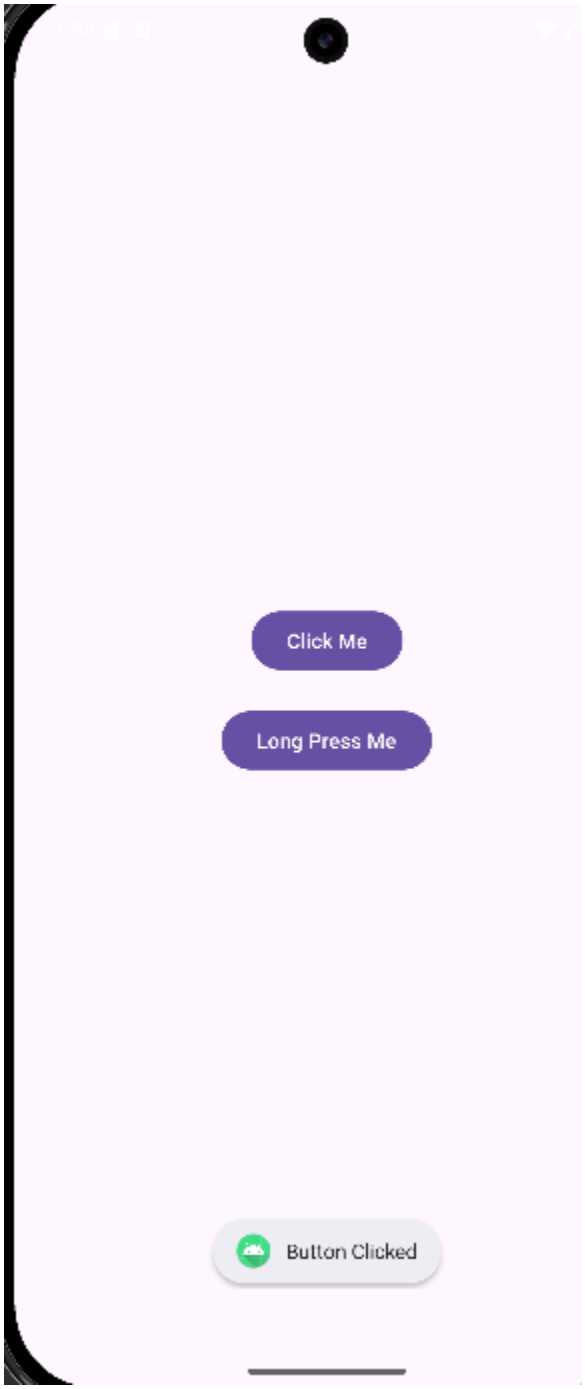
            ).show();

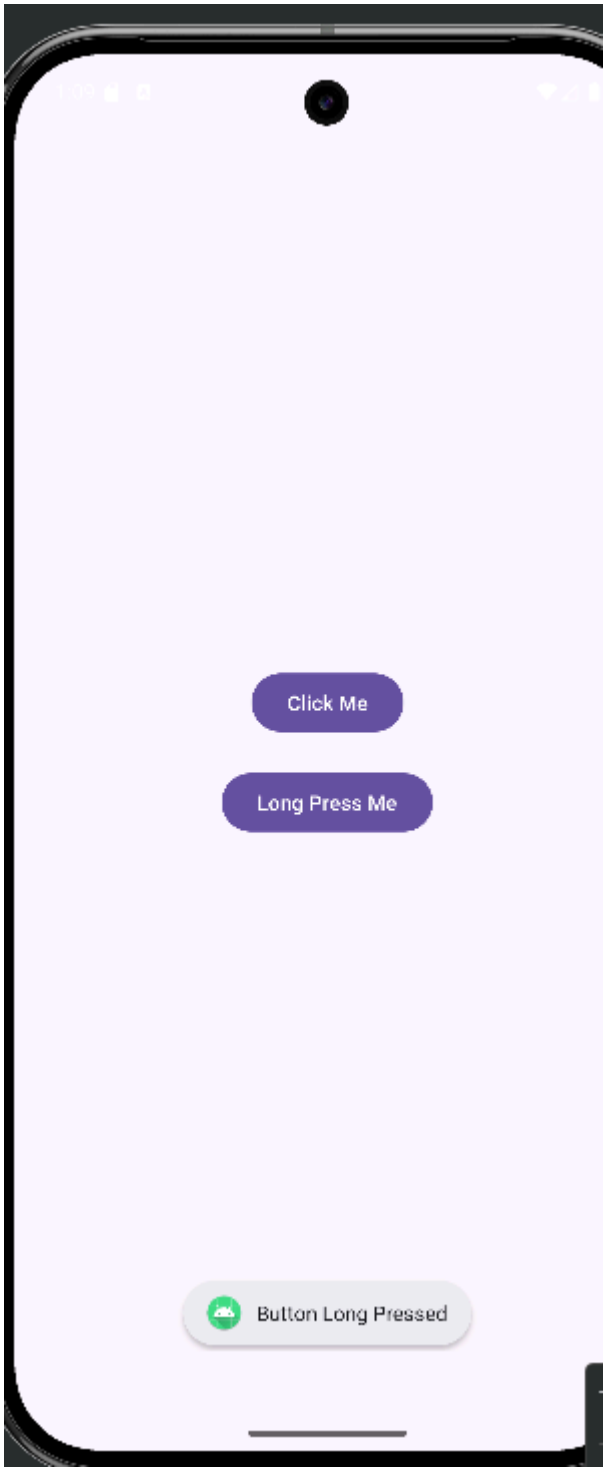
            return true;

        });

    }

}
```


Click Me
Long Press Me
Button Clicked


Click Me
Long Press Me
Button Long Pressed